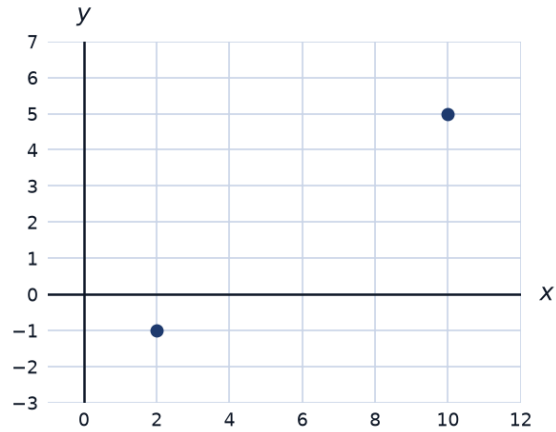


# Coordinate Geometry

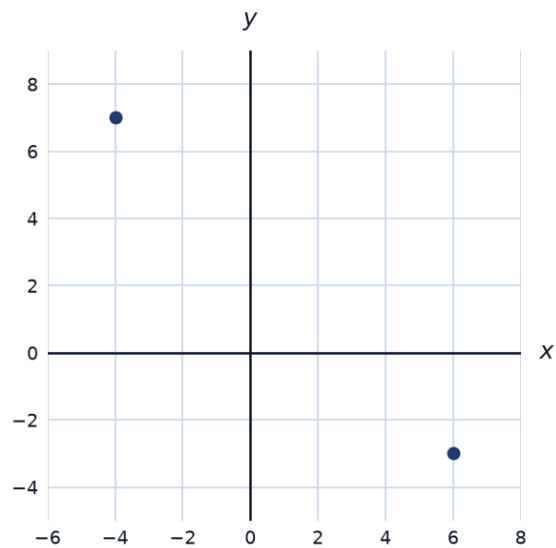


## Distance and midpoint formulas

- 1 Find the distance between  $(2, -1)$  and  $(10, 5)$ .



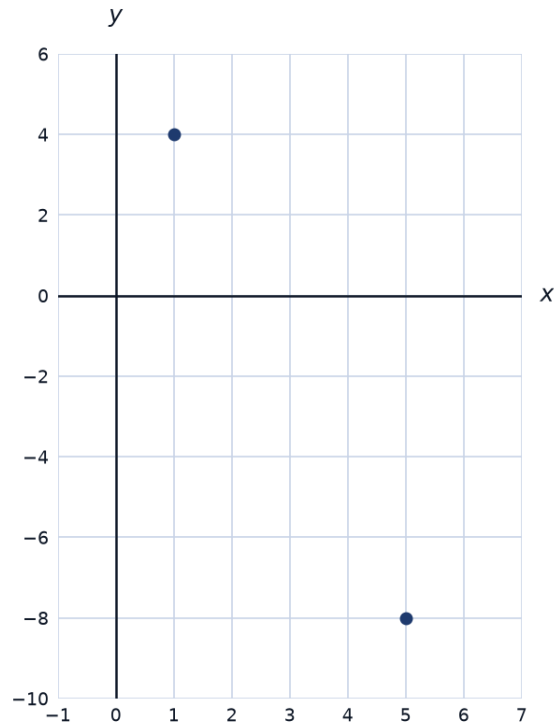
- 2 Find the midpoint of the segment connecting  $(-4, 7)$  and  $(6, -3)$ .



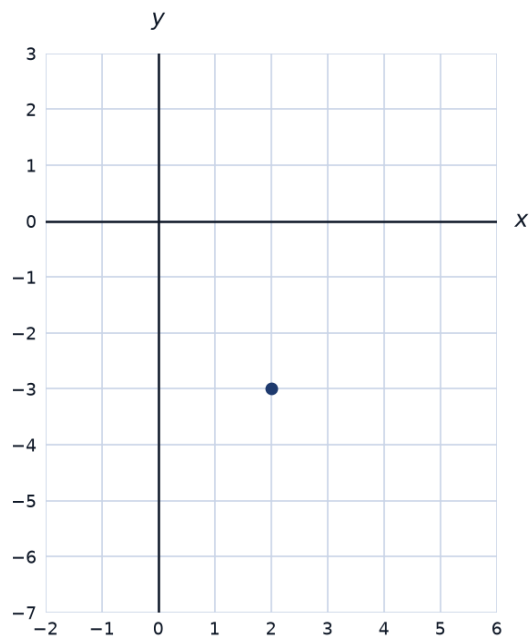
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## Slope and equations of lines

- 3 Find the slope of the line through  $(1, 4)$  and  $(5, -8)$ .



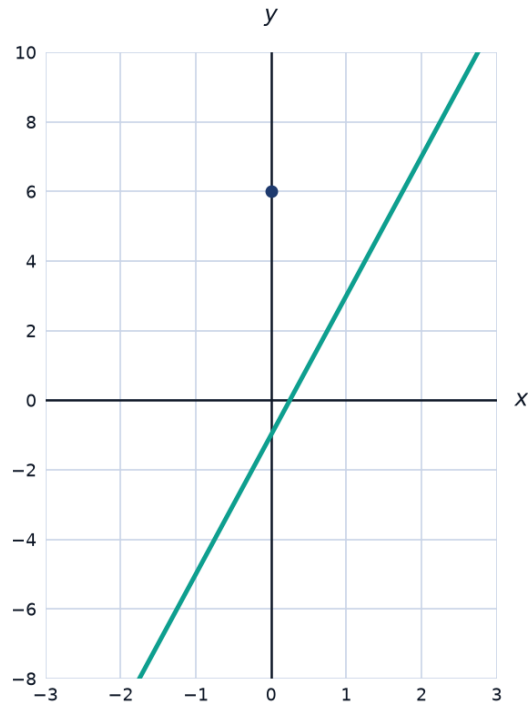
- 4 Write the equation of the line through  $(2, -3)$  with slope  $\frac{1}{2}$ , in slope-intercept form.



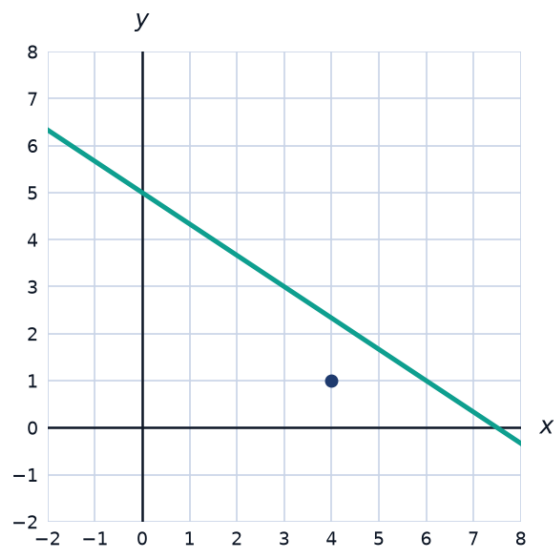
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## Parallel and perpendicular lines

- 5 Find the equation of the line parallel to  $y = 4x - 1$  passing through  $(0, 6)$ .



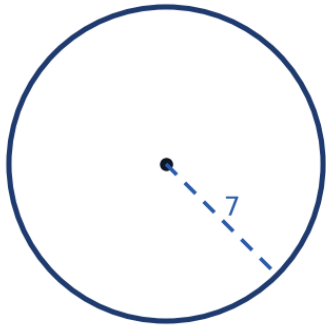
- 6 Find the equation of the line perpendicular to  $y = -\frac{2}{3}x + 5$  passing through  $(4, 1)$ .



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## The equation of a circle

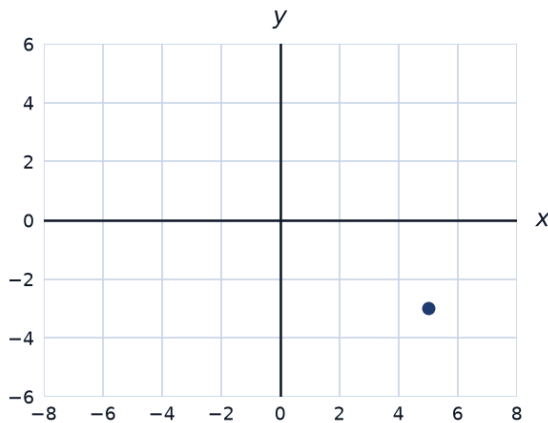
- 7 Write the equation of a circle with center  $(3, -2)$  and radius 7.



- 8 Find the center and radius of the circle  $x^2 + y^2 - 6x + 4y - 3 = 0$ , by completing the square.
- 

### Reflections, translations, and symmetry

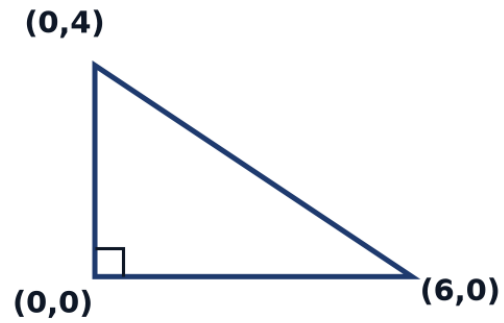
- 9 Find the coordinates of the point  $(5, -3)$  after a reflection over the  $x$ -axis, then after a reflection over the  $y$ -axis.



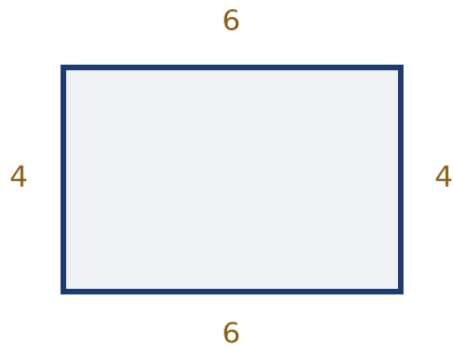
- 10 A point  $(a, b)$  is translated 3 units right and 5 units down. Write the coordinates of the image point.
- 

### Finding area using the coordinate plane

- 11 Find the area of the triangle with vertices  $(0, 0)$ ,  $(6, 0)$ , and  $(0, 4)$ .



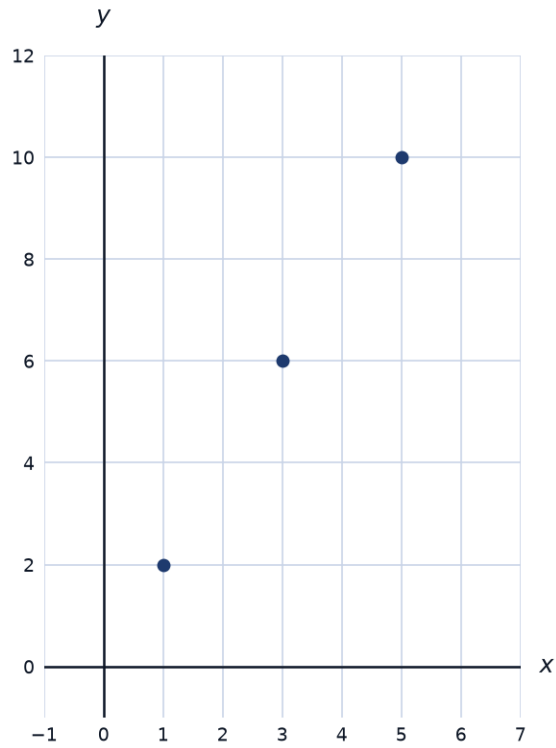
- 12 A rectangle has vertices  $(1, 1)$ ,  $(1, 5)$ ,  $(7, 5)$ , and  $(7, 1)$ . Find its area and perimeter.



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**Determining if points are collinear**

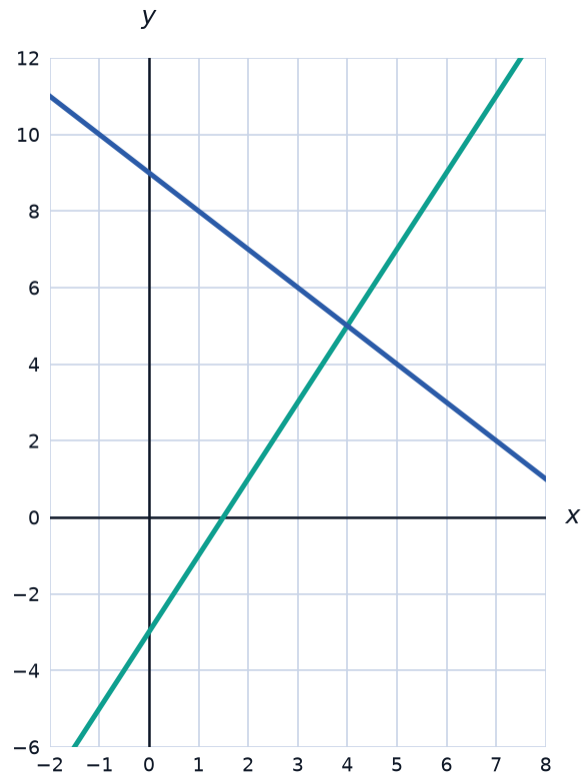
- 13 Determine whether the points  $(1, 2)$ ,  $(3, 6)$ , and  $(5, 10)$  are collinear, by comparing slopes.



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### Finding the intersection of two lines

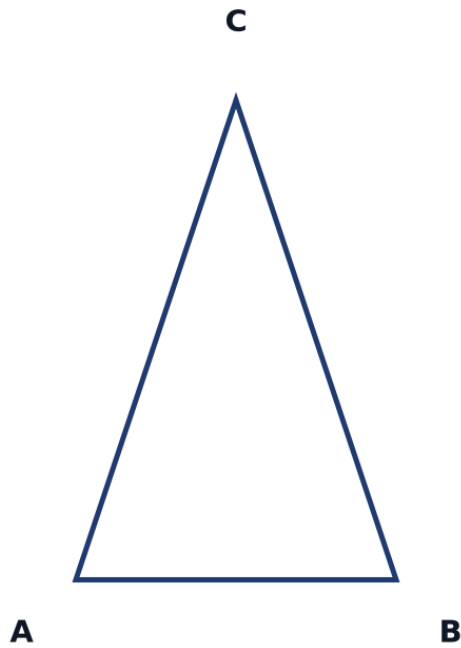
- 14 Find the point of intersection of  $y = 2x - 3$  and  $y = -x + 9$ .



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**Combining coordinate geometry concepts**

- 15** This question has two parts. Triangle  $ABC$  has vertices  $A(0, 0)$ ,  $B(4, 0)$ , and  $C(2, 6)$ . (a) Find the length of each side of the triangle. (b) Determine whether the triangle is isosceles, and justify your answer.

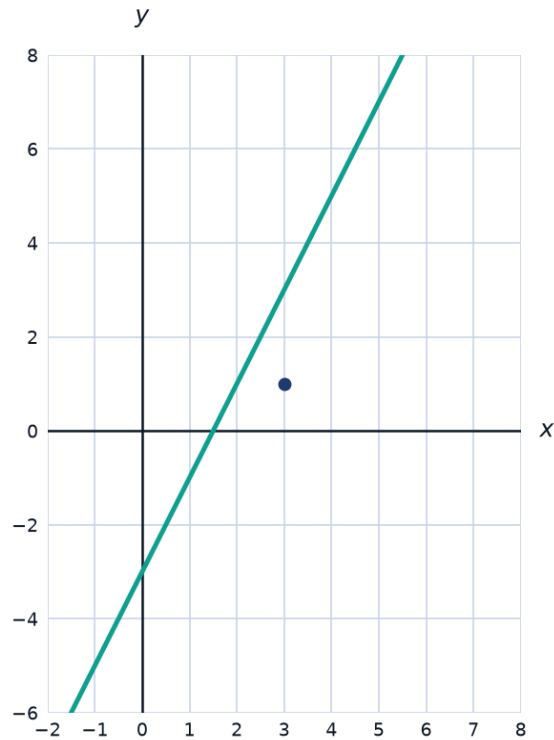


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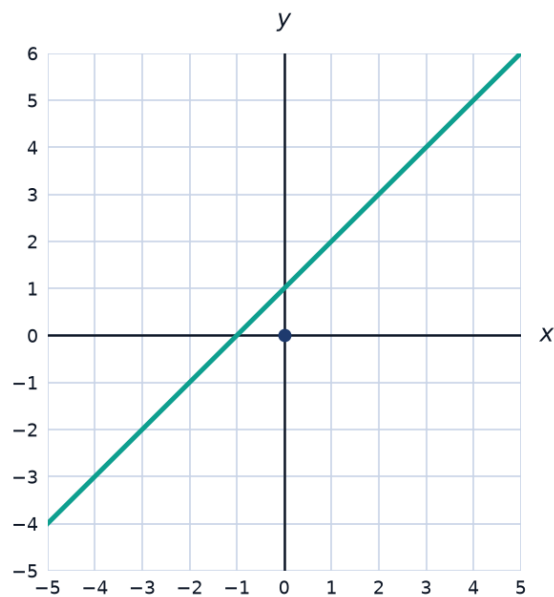
**Graphing inequalities in the coordinate plane**



- 16 Determine whether the point  $(3, 1)$  satisfies the inequality  $y < 2x - 3$ .



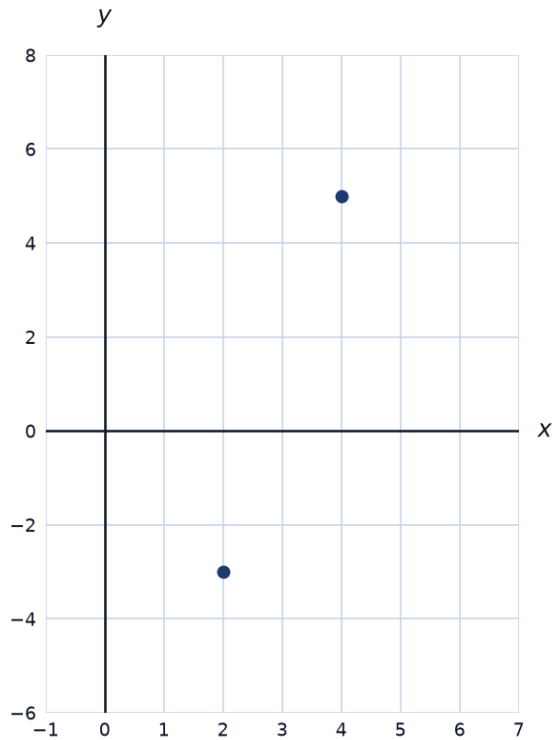
- 17 A region is defined by  $y \geq x + 1$ . Determine whether  $(0, 0)$  lies in this region.



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## The equation of a parabola in the coordinate plane

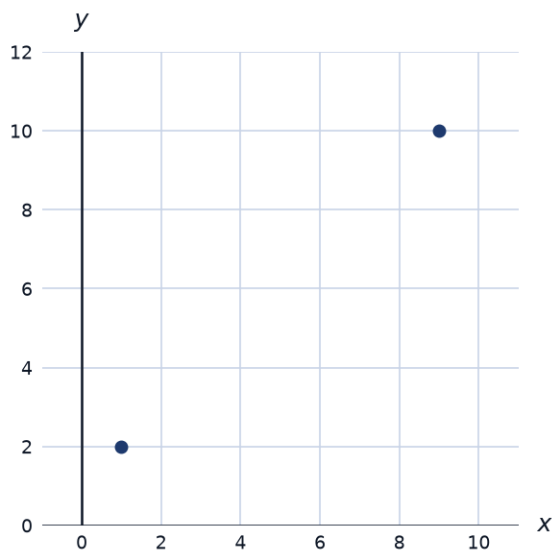
- 18 Write the equation of a parabola with vertex  $(2, -3)$  that passes through  $(4, 5)$ .



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### Finding points that divide a segment in a given ratio

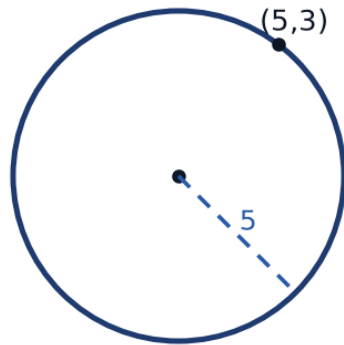
- 19 Find the point that divides the segment from  $(1, 2)$  to  $(9, 10)$  in a ratio of  $1 : 3$  (closer to the first point).



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### A multi-part problem combining lines and circles

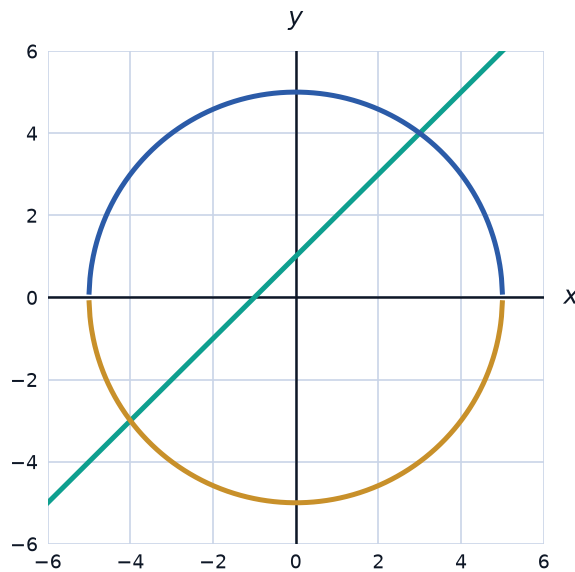
- 20** This question has two parts. A circle has equation  $(x - 2)^2 + (y + 1)^2 = 25$ . (a) State the center and radius. (b) Determine whether the point  $(5, 3)$  lies inside, on, or outside the circle, by computing its distance from the center.



---

### Systems of a line and a circle

- 21** Find the points of intersection of the line  $y = x + 1$  and the circle  $x^2 + y^2 = 25$ .



---

### Multiple choice — easy

- 22 Find the distance between  $(8, 8)$  and  $(18, 32)$ .
- a. 27
  - b. 34
  - c. 25
  - d. 26
- 23 Find the distance between  $(1, -2)$  and  $(-26, 118)$ .
- a. 123
  - b. 122
  - c. 124
  - d. 147
- 24 Find the distance between  $(-2, 7)$  and  $(-29, -29)$ .
- a. 63
  - b. 46
  - c. 45
  - d. 44
- 25 Find the distance between  $(-5, 0)$  and  $(4, 40)$ .
- a. 49
  - b. 42
  - c. 41
  - d. 40

- 26 Find the distance between  $(0, 2)$  and  $(8, 17)$ .
- a. 16
  - b. 18
  - c. 17
  - d. 23
- 27 Find the distance between  $(5, 8)$  and  $(-31, -40)$ .
- a. 60
  - b. 61
  - c. 84
  - d. 59
- 28 Find the distance between  $(-2, 4)$  and  $(5, -20)$ .
- a. 26
  - b. 24
  - c. 31
  - d. 25
- 29 Find the distance between  $(-4, -7)$  and  $(6, -31)$ .
- a. 26
  - b. 34
  - c. 25
  - d. 27

**30** Find the distance between  $(6, -3)$  and  $(-12, -27)$ .

a. 29

b. 42

c. 30

d. 31

**31** Find the distance between  $(1, -7)$  and  $(10, -47)$ .

a. 49

b. 40

c. 42

d. 41

**32** Find the distance between  $(-7, -6)$  and  $(17, -51)$ .

a. 52

b. 50

c. 51

d. 69

**33** Find the distance between  $(-7, -2)$  and  $(-25, -26)$ .

a. 29

b. 30

c. 31

d. 42

- 34 Find the distance between  $(3, -5)$  and  $(-17, 43)$ .
- a. 52
  - b. 53
  - c. 68
  - d. 51
- 35 Find the distance between  $(-8, -6)$  and  $(-3, 6)$ .
- a. 13
  - b. 14
  - c. 17
  - d. 12
- 36 Find the distance between  $(8, -5)$  and  $(-19, -125)$ .
- a. 122
  - b. 123
  - c. 124
  - d. 147

---

### Multiple choice — medium

- 37 Find the midpoint of the segment connecting  $(-9, -10)$  and  $(1, -8)$ .
- a.  $(4, -9)$
  - b.  $(-4, -9)$
  - c.  $(-4, -8)$
  - d.  $(-3, -9)$

- 38 Find the midpoint of the segment connecting  $(9, -5)$  and  $(1, 11)$ .
- a.  $(5, 4)$
  - b.  $(6, 3)$
  - c.  $(5, 3)$
  - d.  $(-5, 3)$
- 39 Find the midpoint of the segment connecting  $(1, 9)$  and  $(-7, 9)$ .
- a.  $(-2, 9)$
  - b.  $(3, 9)$
  - c.  $(-3, 9)$
  - d.  $(-3, 10)$
- 40 Find the midpoint of the segment connecting  $(-6, 0)$  and  $(4, 12)$ .
- a.  $(1, 6)$
  - b.  $(0, 6)$
  - c.  $(-1, 6)$
  - d.  $(-1, 7)$
- 41 Find the midpoint of the segment connecting  $(8, -8)$  and  $(-4, 0)$ .
- a.  $(2, -3)$
  - b.  $(2, -4)$
  - c.  $(3, -4)$
  - d.  $(-2, -4)$



- 42 Find the midpoint of the segment connecting  $(2, 8)$  and  $(10, -4)$ .
- a.  $(6, 3)$
  - b.  $(6, 2)$
  - c.  $(7, 2)$
  - d.  $(-6, 2)$
- 43 Find the midpoint of the segment connecting  $(-3, 1)$  and  $(9, 11)$ .
- a.  $(3, 6)$
  - b.  $(3, 7)$
  - c.  $(4, 6)$
  - d.  $(-3, 6)$
- 44 Find the midpoint of the segment connecting  $(7, 3)$  and  $(-11, -11)$ .
- a.  $(-2, -3)$
  - b.  $(-1, -4)$
  - c.  $(2, -4)$
  - d.  $(-2, -4)$
- 45 Find the midpoint of the segment connecting  $(-7, -5)$  and  $(3, 3)$ .
- a.  $(-2, 0)$
  - b.  $(2, -1)$
  - c.  $(-1, -1)$
  - d.  $(-2, -1)$

- 46 Find the midpoint of the segment connecting  $(8, 1)$  and  $(0, -7)$ .
- a.  $(5, -3)$
  - b.  $(4, -3)$
  - c.  $(4, -2)$
  - d.  $(-4, -3)$
- 47 Find the midpoint of the segment connecting  $(4, 4)$  and  $(-8, -10)$ .
- a.  $(2, -3)$
  - b.  $(-2, -2)$
  - c.  $(-2, -3)$
  - d.  $(-1, -3)$
- 48 Find the midpoint of the segment connecting  $(-7, -3)$  and  $(3, 1)$ .
- a.  $(-2, 0)$
  - b.  $(-2, -1)$
  - c.  $(2, -1)$
  - d.  $(-1, -1)$
- 49 Find the midpoint of the segment connecting  $(2, -5)$  and  $(-2, 5)$ .
- a.  $(1, 0)$
  - b.  $(1, -2)$
  - c.  $(0, 0)$
  - d.  $(0, 1)$

50 Find the midpoint of the segment connecting  $(12, 9)$  and  $(-4, 11)$ .

a.  $(-4, 10)$

b.  $(4, 11)$

c.  $(5, 10)$

d.  $(4, 10)$

51 Find the midpoint of the segment connecting  $(8, 7)$  and  $(-8, -9)$ .

a.  $(1, -1)$

b.  $(0, -1)$

c.  $(0, 0)$

d.  $(-2, 0)$

---

### Multiple choice — hard

52 Find the radius of the circle  $x^2 + y^2 + 16x - 18y + 81 = 0$ , by completing the square.

a. 8

b. 7

c. 64

d. 9

53 Find the radius of the circle  $x^2 + y^2 - 18x - 16y + 141 = 0$ , by completing the square.

a. 3

b. 1

c. 4

d. 2

- 54 Find the radius of the circle  $x^2 + y^2 - 16y + 15 = 0$ , by completing the square.
- a. 49
  - b. 7
  - c. 8
  - d. 6
- 55 Find the radius of the circle  $x^2 + y^2 + 10x - 4y + 4 = 0$ , by completing the square.
- a. 25
  - b. 5
  - c. 6
  - d. 4
- 56 Find the radius of the circle  $x^2 + y^2 - 8x - 16y - 41 = 0$ , by completing the square.
- a. 11
  - b. 121
  - c. 12
  - d. 10
- 57 Find the equation of the line perpendicular to  $y = -\frac{5}{4}x$  passing through  $(0, -3)$ , in slope-intercept form.
- a.  $y = -\frac{4}{5}x - 3$
  - b.  $y = \frac{4}{5}x - 3$
  - c.  $y = -\frac{5}{4}x - 3$
  - d.  $y = \frac{4}{5}x - 2$

- 58 Find the equation of the line perpendicular to  $y = -\frac{1}{4}x$  passing through  $(0, 5)$ , in slope-intercept form.
- a.  $y = -\frac{1}{4}x + 5$
  - b.  $y = 4x + 6$
  - c.  $y = -4x + 5$
  - d.  $y = 4x + 5$
- 59 Find the equation of the line perpendicular to  $y = \frac{1}{5}x$  passing through  $(3, 7)$ , in slope-intercept form.
- a.  $y = 5x + 22$
  - b.  $y = \frac{1}{5}x + 22$
  - c.  $y = -5x + 23$
  - d.  $y = -5x + 22$
- 60 Find the equation of the line perpendicular to  $y = -\frac{2}{3}x$  passing through  $(3, -1)$ , in slope-intercept form.
- a.  $y = \frac{3}{2}x - \frac{9}{2}$
  - b.  $y = -\frac{2}{3}x - \frac{11}{2}$
  - c.  $y = -\frac{3}{2}x - \frac{11}{2}$
  - d.  $y = \frac{3}{2}x - \frac{11}{2}$
- 61 Find the equation of the line perpendicular to  $y = \frac{5}{4}x$  passing through  $(7, -3)$ , in slope-intercept form.
- a.  $y = \frac{4}{5}x + \frac{13}{5}$
  - b.  $y = -\frac{4}{5}x + \frac{18}{5}$
  - c.  $y = -\frac{4}{5}x + \frac{13}{5}$
  - d.  $y = \frac{5}{4}x + \frac{13}{5}$

62 Find the point of intersection of  $y = 3x - 17$  and  $y = -2x - 2$ .

- a.  $(3, -8)$
- b.  $(4, -8)$
- c.  $(-3, -8)$
- d.  $(3, 8)$

63 Find the point of intersection of  $y = -x + 5$  and  $y = -2x + 1$ .

- a.  $(4, 9)$
- b.  $(-4, 9)$
- c.  $(-3, 9)$
- d.  $(-4, -9)$

64 Find the point of intersection of  $y = x - 12$  and  $y = 4x - 39$ .

- a.  $(9, 3)$
- b.  $(-9, -3)$
- c.  $(9, -3)$
- d.  $(10, -3)$

65 Find the point of intersection of  $y = -x$  and  $y = 2x - 6$ .

- a.  $(2, 2)$
- b.  $(3, -2)$
- c.  $(2, -2)$
- d.  $(-2, -2)$

66 Find the point of intersection of  $y = 5x + 38$  and  $y = -3x - 10$ .

a.  $(-5, 8)$

b.  $(6, 8)$

c.  $(-6, -8)$

d.  $(-6, 8)$